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Lab Report - 01

Course code: CSE3206

Course Title: Computer Networks

Section: 04

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Introduction :

This report presents the design and implementation of two different network topologies using Cisco Packet Tracer. The objective of this task is to understand basic network design concepts, device interconnections, and data transmission within a Local Area Network (LAN). The first part focuses on a topology with multiple switches, servers, and end users, while the second part demonstrates a LAN topology using a router, switches, and end devices. No router configuration was required as per the task instructions.

Task -1 : Design a network topology with 3 switches each having 1 server and 2 end users. Provide evidence of data transmissions.

Sketch :

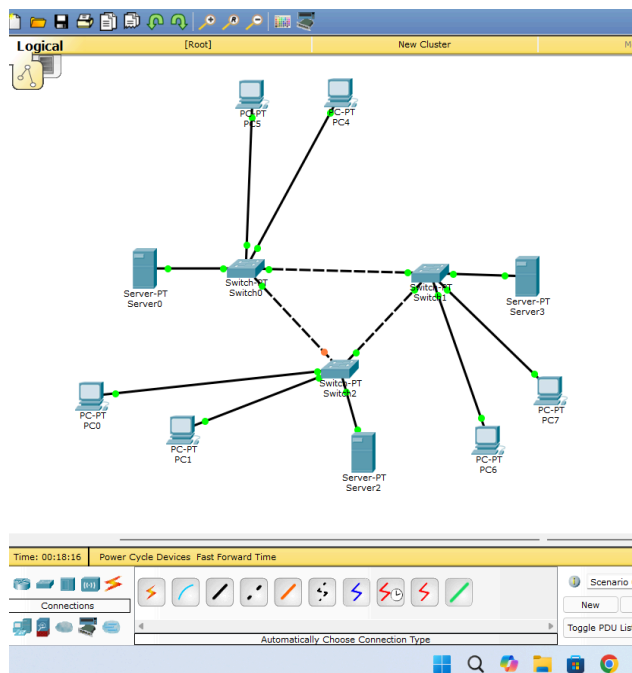
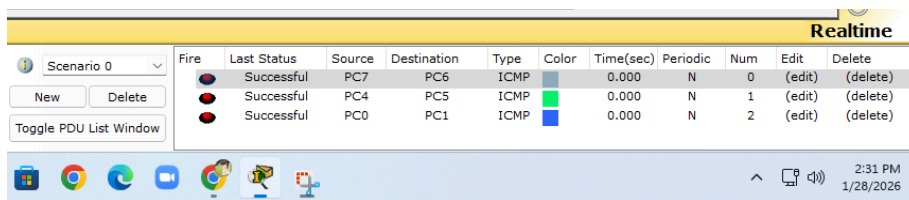


Figure-01 : Network topology with 3 switches each having 1 server and 2 end users

Description :

This network topology consists of three switches, where each switch is connected to one server and two end-user devices. The **switches** are interconnected to allow communication across the entire network through **Copper Cross-Over** wire and the **rest of the devices** were connected using **Copper Straight-Through** cables and assigned IP addresses within the same network. Data transmission was verified by successfully sending ping requests between PCs and servers connected to different switches, confirming proper connectivity and communication.

Evidence of Data Transmissions :



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC7	PC6	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC4	PC5	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	2	(edit)	(delete)

Task -2 : Design a Local Area Network (LAN) topology using 1 router, 2 switches, and 4 end devices.

Sketch :

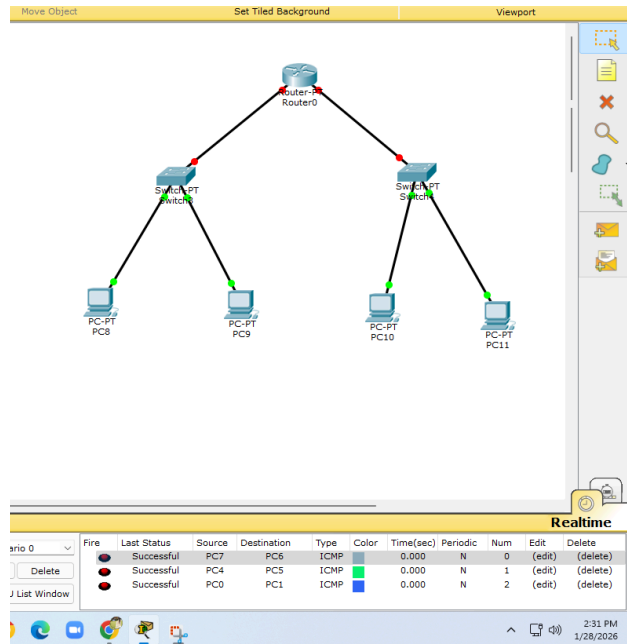


Figure-02 : Local Area Network (LAN) topology using 1 router, 2 switches, and 4 end devices

Description :

This design represents a basic Local Area Network (LAN) consisting of one router, two switches, and four end devices. The router connects the two switches, and each switch connects to two end-user devices. Although the router was not configured, the logical topology clearly demonstrates how routers and switches are used to structure a LAN. The design helps illustrate basic network organization and device interconnections. Here we used only Copper Straight-Through wires because all are connected with different devices .

Task -3 : Connectivity wiring

Table :

SL No.	Connectivity	Wiring Type
1	Switch---Router	Copper Straight-Through
2	Router---Router	Copper Cross-Over
3	Hub---PC3	Copper Straight-Through

4	PC---Router	Copper Straight-Through
5	PC1---PC4	Copper Cross-Over

Conclusion :

In this task, multiple network topologies were successfully designed and implemented using Cisco Packet Tracer. The first design demonstrated effective communication between servers and end users across three interconnected switches, with data transmission verified through successful ping tests. The second design presented a simple Local Area Network (LAN) using one router, two switches, and four end devices, with all devices and connections clearly labeled in the logical topology diagram. Additionally, proper connectivity wiring was completed to ensure correct physical connections between all network components. Overall, this activity enhanced practical understanding of network design, device interconnection, and basic data communication within a LAN.

